



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

CORIZO — ARTIFICIAL INTELLIGENCE INTERNSHIP

Cardiovascular Disease Prediction & Spotify Songs' Genre Segmentation

A Machine Learning Internship Presentation Report

INTERNSHIP DURATION

05 December 2025 – 05 January 2026 (1 Month)

STUDENT CREDENTIALS

ARHAM JAIN

Enrolment / Roll No.

25SCS1003001426

Program & Section

**B.Tech CSE-AI/ML,
2CSE-22**

HOST ORGANIZATION

Corizo

Industry-Mentored AI Internship Program

Candidate & Program Overview



Candidate Details

Student Name: Arham Jain

Roll Number: 25SCS1003001426

Degree: B.Tech CSE – AI/ML (IBM-ICE)

Section: 2CSE-22

Academic Batch: 2025-2029

University: IILM University, Greater Noida, U.P.



Internship Profile

Program Name: Corizo — Artificial Intelligence Internship

Projects Undertaken: 2 (Cardiovascular Disease Prediction; Spotify Genre Segmentation)

Duration: 1 Month (05 Dec 2025 – 05 Jan 2026)

Corizo Dice ID: CRZ140452

Academic Head: Hemant Ingle

Organization Profile: Corizo

Corizo is an Indian ed-tech platform that provides internships, professional training programs, career guidance and mentorship to engineering and management students, aiming to bridge the gap between classroom learning and industry requirements.



Mentorship Model: Pairs learners with experienced industry mentors through a structured, project-based curriculum.



Multi-Domain Training: Offers programs across Data Science, Machine Learning, Web Development, Cyber Security and Cloud Computing.



Verified Certification: Issues an internship completion certificate on successful evaluation of assigned project work.

2

MACHINE LEARNING PROJECTS COMPLETED

96,965

TOTAL DATASET RECORDS ANALYZED

22

VISUALIZATIONS & CHARTS PRODUCED

Internship Objectives & Approach



Data Preprocessing

Clean, validate and engineer features across two real-world datasets of over 96,000 combined records.



Exploratory Analysis

Visualize distributions, relationships and correlations to understand each dataset before modeling.



Model Building

Train five supervised classifiers for disease prediction, and apply K-Means clustering for genre segmentation.



Evaluation & Insight

Compare models on accuracy, precision, recall and F1-score; interpret clusters against real genre labels.

Technologies & Tools Used

Language & Data

- Python 3
- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn

Machine Learning

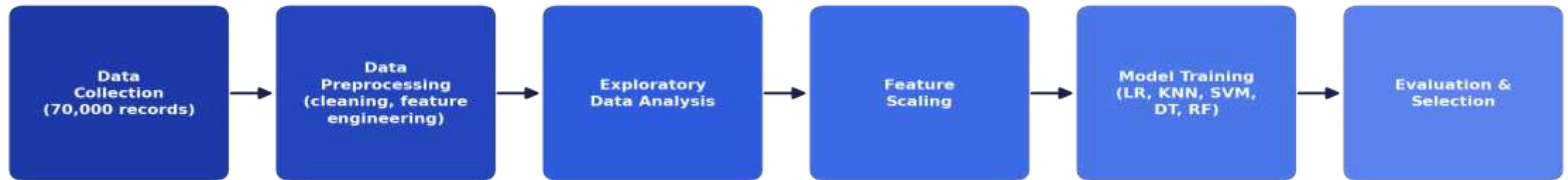
- Scikit-learn
- K-Means · PCA
- 5 classifiers

Environment & Source

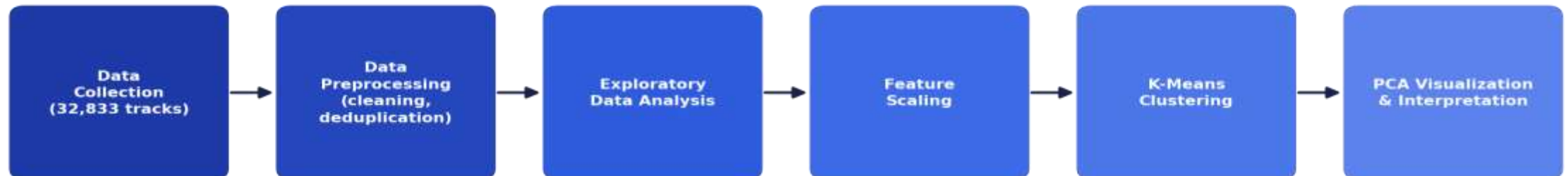
- Jupyter Notebook
- Kaggle datasets

System Architecture

Project 1 — Cardiovascular Disease Prediction (Supervised Pipeline)



Project 2 — Spotify Songs' Genre Segmentation (Unsupervised Pipeline)



Project 1: Cardiovascular Disease Prediction



The Health Challenge

Cardiovascular disease is the leading cause of death worldwide — the WHO estimates 19.8 million deaths in 2022, close to a third of all deaths globally. Many risk factors (blood pressure, cholesterol, glucose, lifestyle habits) are measurable well before a cardiac event, motivating data-driven early screening from routine health-check data.



The Dataset

70,000 anonymized patient records (68,613 after cleaning)

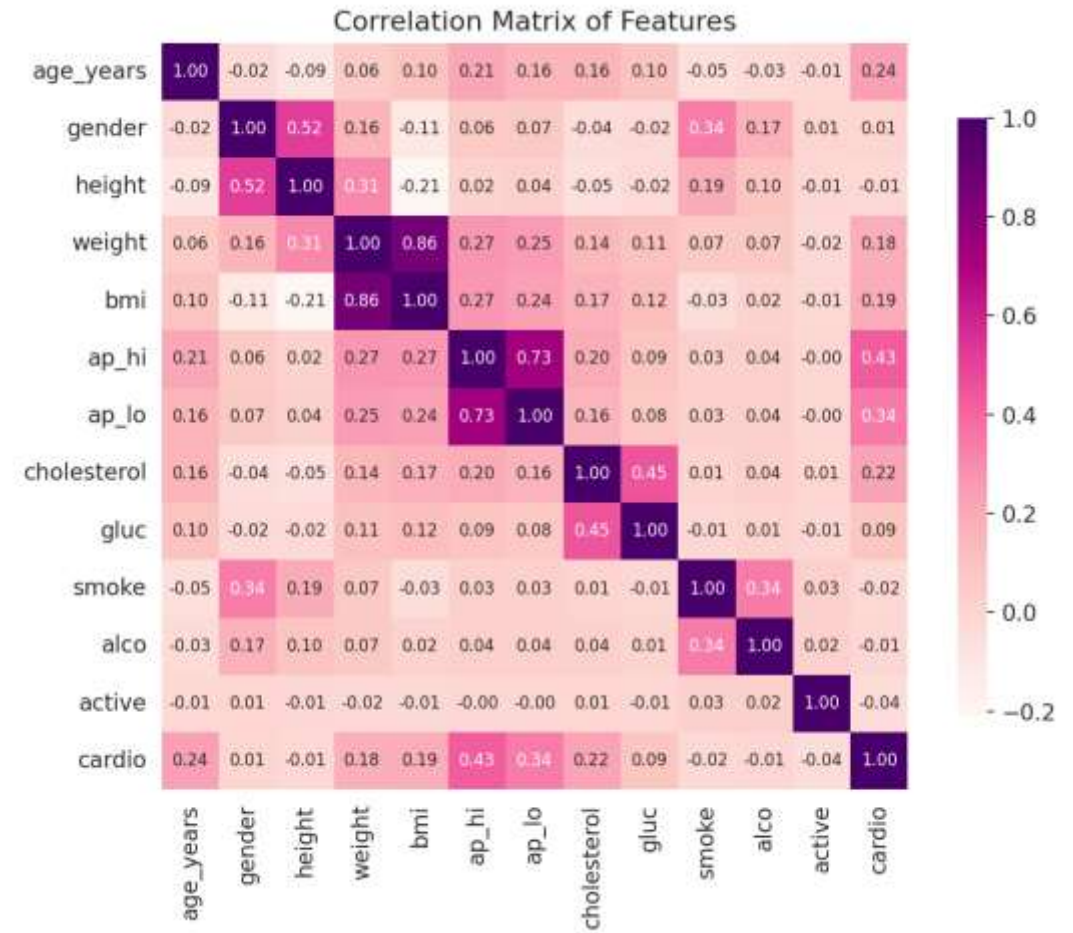
11 clinical & lifestyle attributes: age, gender, height, weight, blood pressure, cholesterol, glucose, smoking, alcohol, activity

2 engineered features: age in years, BMI

Binary target: presence / absence of cardiovascular disease (near-perfectly balanced, ~50/50)

Cardiovascular Disease — Exploratory Findings

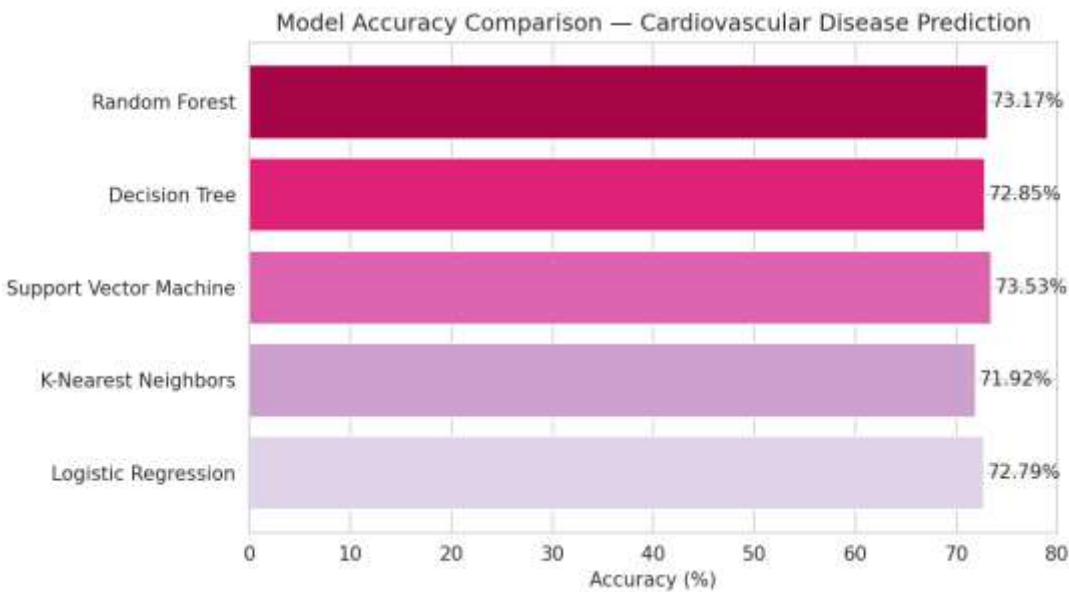
- Disease prevalence rises steadily with age and BMI, and sharply with cholesterol — from 43.5% (normal) to 76.2% (well above normal).
- Systolic BP (0.43), diastolic BP (0.34), age (0.24) and cholesterol (0.22) are the attributes most correlated with disease.
- Smoking, alcohol and activity show only a weak relationship in the raw data — most likely self-report bias rather than a true protective effect.



Cardiovascular Disease — Model Performance

Model	Accuracy	Precision	F1-score
Logistic Regression	72.79%	75.26%	70.91%
K-Nearest Neighbors	71.92%	72.73%	70.90%
Support Vector Machine ★	73.53%	76.07%	71.71%
Decision Tree	72.85%	73.19%	72.18%
Random Forest	73.17%	75.45%	71.44%

Support Vector Machine was the best performer at 73.53% accuracy. All five models cluster within a narrow 72–74% band — the ceiling here is set more by overlap in the clinical data itself than by algorithm choice.



Project 2: Spotify Songs' Genre Segmentation



The Music Challenge

Streaming platforms curate music using numerical audio features (danceability, energy, valence, tempo) as much as explicit choices. Genre is a human, cultural label layered on a continuous acoustic space — two songs in the same genre can sound very different. This project explores whether audio features alone can recover meaningful genre-like groupings.



The Dataset

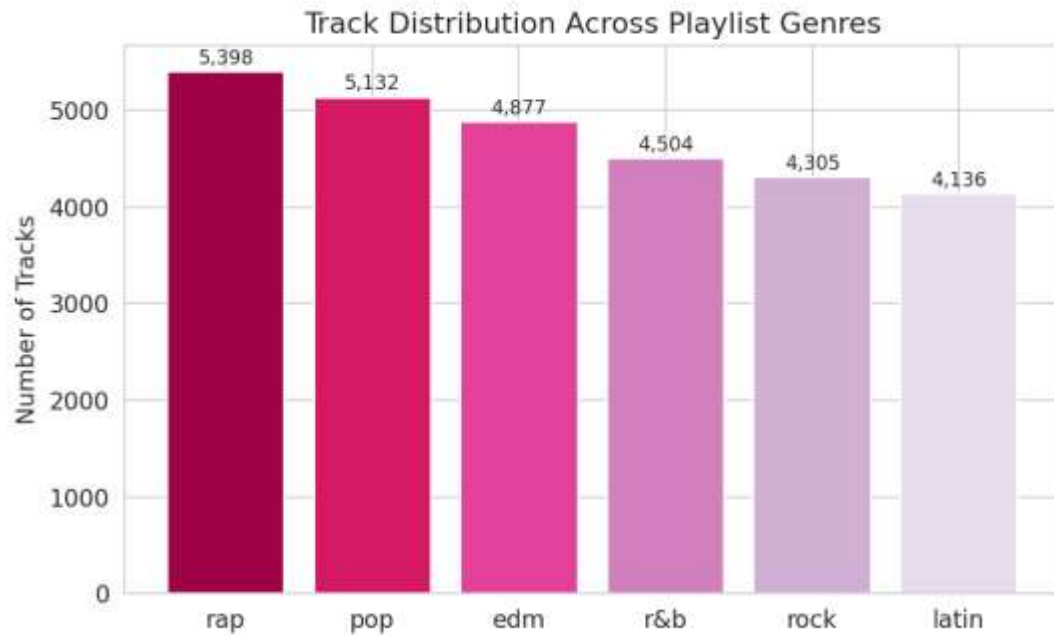
32,833 Spotify tracks (28,352 unique after de-duplication)

6 playlist genres: pop, rap, rock, latin, r&b, edm — across 24 subgenres

11 numerical audio features used: danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo

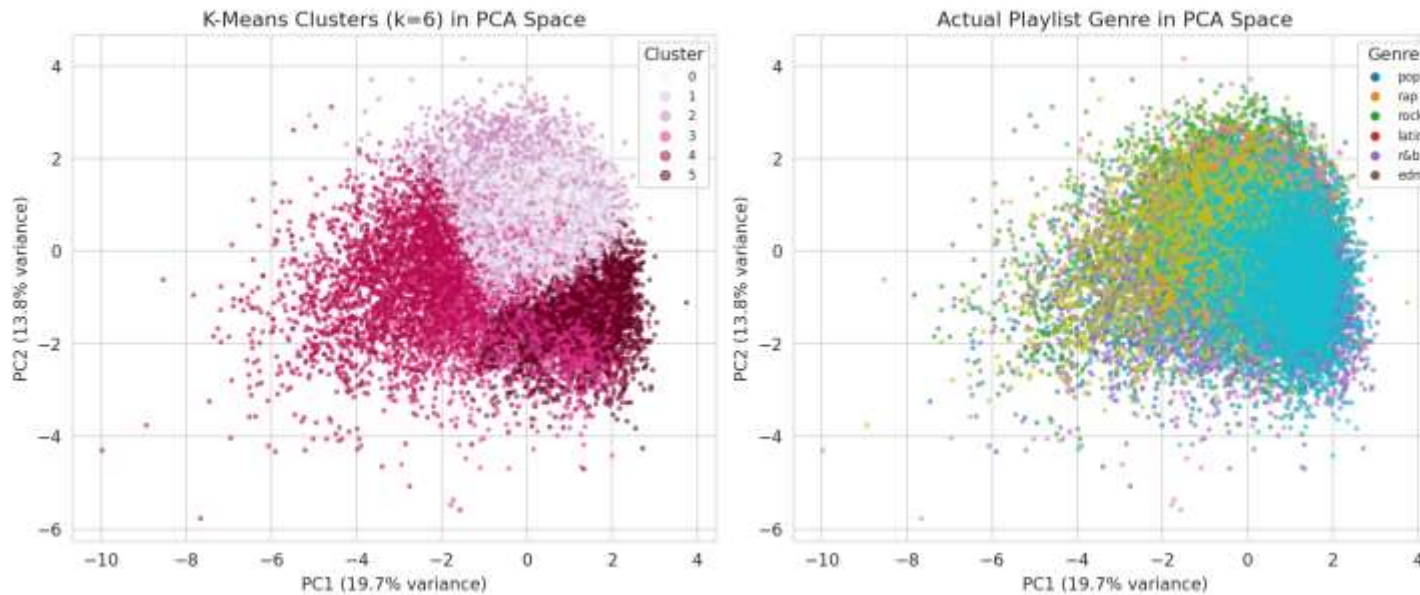
Unsupervised task — genre label used only to validate clusters afterward

Spotify — Exploratory Findings



- The six genres are fairly evenly represented (4,136–5,398 tracks each), so no single genre dominates the clustering by sample size alone.
- EDM has the highest energy and lowest acousticness; rock has the lowest danceability; latin has the highest valence.
- Energy and loudness are strongly positively correlated; energy and acousticness are strongly negatively correlated.

Spotify — Clustering Results



k = 6

CLUSTERS CHOSEN

0.114

SILHOUETTE SCORE

Rap (58.0%) and EDM (57.7%) form the purest clusters — each has a distinctive acoustic signature (high speechiness for rap; high energy, low acousticness for EDM) that K-Means isolates without ever seeing the genre label.

Pop, latin, r&b and rock overlap much more (21–32% purity) — audio features alone don't cleanly separate these genres.

Key Outcomes & Skills Demonstrated

- ✓ **End-to-End Data Pipelines:** Cleaned and engineered two real-world datasets totalling nearly 97,000 records, from raw CSV to analysis-ready form.
- ✓ **Supervised Classification:** Trained and compared 5 algorithms (Logistic Regression, KNN, SVM, Decision Tree, Random Forest), reaching 73.53% accuracy with SVM.
- ✓ **Unsupervised Clustering:** Applied K-Means and PCA to segment 28,352 tracks by audio features, and validated clusters against real genre labels.
- ✓ **Statistical & Visual Storytelling:** Produced 22 charts — correlation heatmaps, cluster plots, confusion matrices — to communicate findings clearly.
- ✓ **Critical, Honest Evaluation:** Reported counter-intuitive and weak results (e.g. smoking vs. disease rate, low silhouette score) rather than only favourable ones.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	Artificial Intelligence Internship
Issuing Organization	Corizo
Duration	05 December 2025 – 05 January 2026 (1 Month)
Corizo Dice ID	CRZ140452
Academic Head	Hemant Ingle
Certifying Partner	IIT Bombay’s Mood Indigo

Internship Completion Certificate



Thank You!



Arham Jain

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**School of Computer Science &
Engineering**

IILM University, Greater Noida, U.P.